

No. of Functional Features Included in the Solution

Field	Value
Date	05 July 2026
Team ID	(Your Team ID)
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Registration & Login	Allows farmers to securely access the application.	Authentication	Done	High
2	Soil Data Input	Collects soil parameters such as N, P, K, pH, and moisture.	Data Collection	Done	High
3	Weather Data Input	Accepts temperature, humidity, and rainfall information.	Data Collection	Done	High
4	Crop Recommendation	Suggests the most suitable crop using ML algorithms.	Recommendation Engine	Done	High
5	Fertilizer Recommendation	Recommends appropriate fertilizers based on soil conditions.	Recommendation Engine	Done	High
6	Yield Prediction	Predicts expected crop yield before cultivation.	Prediction Module	Done	High
7	Dashboard & Reports	Displays recommendations and	User Interface	Done	Medium

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
8	Future Weather API Integration	prediction results in a user-friendly dashboard.			
		Automatic retrieval of weather information from online services.	API Integration	Pending	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Login Page, Dashboard
2	Backend / Logic	3	Crop Recommendation, Fertilizer Recommendation, Yield Prediction
3	Database / Storage	1	Farmer & Input Data Storage
4	API / Integration	1	Weather API (<i>Planned</i>)
5	Security / Authentication	1	User Login & Authentication